



(14-05-2021)

End Term (Even) Semester Regular/Back Examination May 2026

Roll no... 2592029

Name of the Program and semester: BCA/BSc.IT II
Name of the Course : Introduction to Object Oriented Programming
Course Code: TBC 202/TBI 202

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

- a. Explain the main features of Object-Oriented Programming (OOP) with examples.CO1
- b. Describe different control structures in C++ with suitable syntax and examples.CO2
- c. Write a program to implement different types of pointer arithmetic.CO2

Q2. (2X10=20 Marks)

- a. What is access specifier in c++. Explain its type. Also write a program to demonstrate how to access private and protected data members of class outside the class.CO2
- b. Explain early binding and late binding in C++. Also write a program to demonstrate both early and late binding .CO3
- c. Write a program to store and display student details (roll number, name, marks) using a structure.CO1

Q3. (2X10=20 Marks)

- a. What are constructors and destructors? Explain different types of constructors with examples.CO3
- b. What is an abstract class in c++. Explain its feature and importance in OOP. Also write a program to implement abstract class . CO4
- c. Why we need operator overloading in c++. Write a program in C++ to demonstrate overloading of binary multiplication operator.CO4

Q4. (2X10=20 Marks)

- a. Explain types of inheritance in C++ with diagrams and syntax. Also write a program to demonstrate how to solve ambiguity problem in multiple inheritance using virtual class.CO5
- b. Write short note on exception handling in c++. Also write a program to demonstrate the working of try, multiple catch statements , and throw statement.CO5
- c. Explain the following with example:CO3
 - i. Static variable
 - ii. Instance variable
 - iii. Reference variable

Q5. (2X10=20 Marks)

- a. Explain file handling in C++. Also explain file modes and basic operations of file. Also write a program to perform writing and reading to/from file.CO4
- b. Write a note on friend function and friend class in C++. Write a program to implement friend class.CO3
- c. Write program to create a class Student which has data member id and name, define overloaded constructor to initialize object and copy constructor to initialize one object by another existing object keep name of parameter of parameterized constructor same as data members. Test function of the program in main function.CO5